

Kickstarter GO-Threshold Calibration Study

Recalibrating the pipeline's success-prediction threshold for the Kickstarter domain, validated with a proper train/test split (5-fold cross-validation) rather than in-sample tuning -- plus a bootstrap-checked test of whether a better composite score can beat the production score outright.

GENERATED

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DATASET

141 Kickstarter proposals

VALIDATION

Stratified 5-fold cross-validation

0.764

ROC-AUC, PRODUCTION SCORE

0%

RECALL, BEFORE CALIBRATION

58%

RECALL, AFTER (HELD-OUT)

26%

PRECISION, AFTER (HELD-OUT)

0.773

ROC-AUC, HONEST COMPOSITE (NESTED)

What is this report, in plain terms?

In the earlier Kickstarter validation, the pipeline's GO/HOLD/NO-GO decision was made using a threshold (score ≥ 0.725 OR confidence ≥ 0.811) that had been calibrated on a completely different dataset — 12 biotech/semiconductor proposals. That threshold never fired "GO" on any of the 141 Kickstarter campaigns, even ones that actually succeeded, because Kickstarter proposals structurally score lower on this rubric than VC-style proposals do (explained in the previous report).

This report asks a different, more useful question: **if we pick a decision threshold that's actually appropriate for the Kickstarter score distribution, does the AI's score have real power to separate winners from losers?** To answer that honestly, we can't just search for the "best" threshold and then grade ourselves on the same data we used to find it — that guarantees a flattering but meaningless number. Instead we use **train/test evaluation**: fit the threshold on part of the data, and check it on a different part it never saw. This report walks through exactly how that was done, and what it found.

1. Methodology: How the Threshold Was Calibrated and Tested

The decision rule being calibrated

The pipeline's verdict rule has always had this shape: predict a successful proposal ("GO") if **score $\geq \tau_r$ OR confidence $\geq \tau_c$** . The production system uses $\tau_r=0.725$ and $\tau_c=0.811$, calibrated on the 12-proposal biotech dataset. This study searches for better values of τ_r and τ_c specifically for Kickstarter, keeping the same OR-logic shape so it's a direct, like-for-like recalibration rather than a new mechanism.

Why train/test split, not just "pick the best threshold and report it"

In simple terms: if you try every possible threshold on all 141 proposals and report whichever one scores best on those same 141 proposals, you are guaranteed to get a good-looking number — but it tells you nothing about whether that threshold will work on the *next* Kickstarter campaign it hasn't seen. That's called overfitting. The fix is to test the threshold on data it was not calibrated on.

With only 141 proposals and just 24 real successes, a single 70/30 train/test split would leave as few as 7 successes in the test set — too few to trust a precision/recall estimate. Instead we use **stratified 5-fold cross-validation**: the 141 proposals are split into 5 roughly equal groups ("folds"), each containing a proportional share of successes (~4-5 per fold, matching the 17% overall success rate). We then repeat the following 5 times, once per fold:

1. Hold one fold out as the **test set** (never touched during calibration).
2. Use the remaining 4 folds (the **training set**) to grid-search for the τ_r, τ_c pair that maximizes F1 score — not accuracy (explained below).
3. Apply that threshold to the held-out test fold and record how it did.

Every one of the 141 proposals ends up in exactly one test fold, evaluated using a threshold it never influenced. The 5 test-fold results are then combined into one overall confusion matrix — this is the honest, out-of-sample "after calibration" result in this report.



Illustration of round 1 of 5: fold 1 held out as test, folds 2-5 used to fit the threshold. This repeats 5 times, rotating which fold is held out, so every proposal is tested exactly once.

Why F1 instead of accuracy as the calibration objective

Only 17% of Kickstarter campaigns in this dataset actually succeeded. A model that predicts "will fail" for every single proposal already scores 83% accuracy without ever identifying a real success — which is exactly the failure mode of the original transferred threshold. **F1 score** (the balance of precision and recall) forces the calibration to actually find successes, not just avoid false alarms, so it was used as the objective for selecting τ_r and τ_c on each training fold.

An additional, threshold-independent check

Before even picking a threshold, we also computed **ROC-AUC** and **Average Precision** directly from the raw AI score, with no threshold involved at all. These answer a more fundamental question: does the score contain real information about funding success, regardless of where any cutoff is drawn?

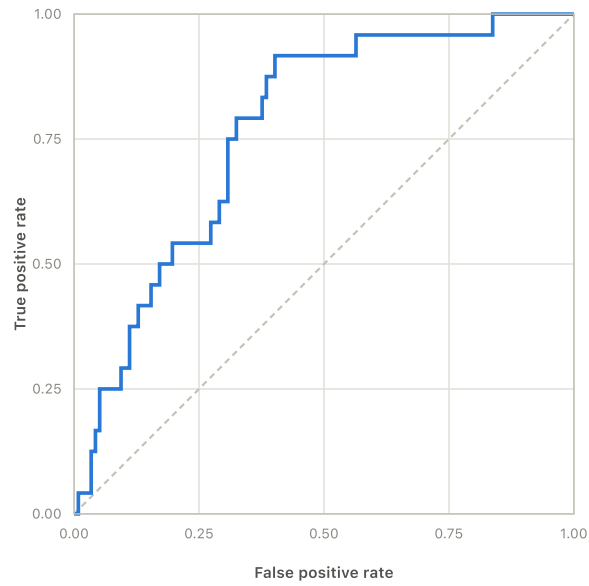
2. Does the Score Have Real Signal? (Threshold-Free Check)

0.764

ROC-AUC
0.5 = random, 1.0 = perfect

0.361

AVERAGE PRECISION
vs. 17% random baseline



In simple terms: ROC-AUC of 0.764 means that if you picked one successful campaign and one failed campaign at random, the AI would rank the successful one higher **76% of the time**. 0.5 would mean the score is no better than a coin flip; 0.764 is a real, meaningful signal (a widely used rule of thumb treats 0.7–0.8 as "fair to good" discrimination). Average precision of 0.361 is roughly **2.1×** better than randomly guessing, given that only 17% of campaigns actually succeed. This confirms the earlier Spearman correlation finding from a different angle: the score is informative — the earlier 0% recall result was a threshold-placement problem, not a sign that the score itself carries no information.

3. Before vs. After Calibration

All three rows below use the exact same underlying AI scores and the exact same OR-logic decision rule — only the threshold values differ.

Scenario	τ_r (score)	τ_c (confidence)	TP	FP	FN	TN	Accuracy	Precision	Recall	F1
Before (production threshold, transferred)	0.725	0.811	0	0	24	117	83.0%	—	0.0%	—
In-sample calibration (overfitting reference)	0.5505	0.6475	22	46	2	71	66.0%	32.4%	91.7%	0.4782
Cross-validated (held-out, honest result)	varies/fold	varies/fold	14	39	10	78	65.2%	26.4%	58.3%	0.3637

Before (production threshold, unmodified): 0% recall — confirmed in the prior report.

In-sample (threshold fit and tested on the same 141 proposals): looks great (91.7% recall, F1=0.478) but this number is **not trustworthy** — it's graded on the data it was tuned to fit.

Cross-validated / held-out (the honest result): recall rises from 0% to **58.3%**, precision lands at **26.4%** (vs. a 17.0% random baseline — a real, if modest, lift), and F1 improves from undefined to **0.364**. This is the number that should be quoted as the pipeline's actual, generalizable performance on Kickstarter-style proposals.

4. Per-Fold Detail: How Stable Is the Calibrated Threshold?

The table below shows what threshold was chosen in each of the 5 training rounds, and how it performed on that round's held-out test fold. This is useful transparency: it shows whether the "right" threshold is consistent across different subsets of the data, or whether it's jumping around (a sign of an unstable estimate given the small number of successes).

Fold	Test n	Test successes	τ_r fit on train	τ_c fit on train	Train F1	Held-out F1	Held-out precision	Held-out recall
Fold 1	29	5	0.5605	0.6485	0.4848	0.375	27%	60%
Fold 2	29	5	0.607	0.6435	0.5	0.0	0%	0%
Fold 3	28	5	0.5505	0.656	0.4865	0.375	27%	60%
Fold 4	28	5	0.5605	0.6425	0.4638	0.4706	33%	80%
Fold 5	27	4	0.5505	0.6465	0.4931	0.4	25%	100%

In simple terms: τ_r (the score cutoff) stays tight across all 5 folds (0.55–0.61), which is reassuring — it means the "right" score cutoff isn't just an artifact of which proposals happened to be in the training set. Held-out F1 per fold is noisier (0.0 to 0.47), which is expected with only ~5 successes per test fold — a single missed or extra success swings the fold's F1 a lot. This is exactly why the **aggregated** result (pooling all 5 test folds into one confusion matrix, Section 3) is the number to trust, not any individual fold.

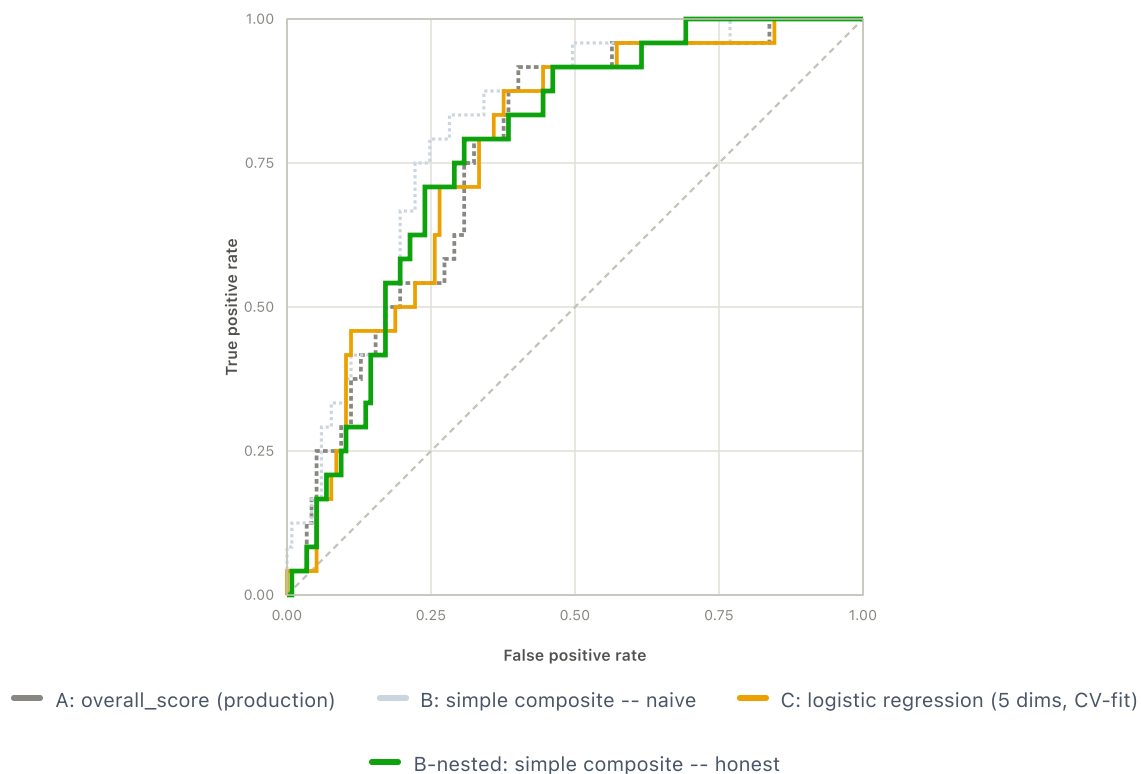
5. Improving the Score: Can a Better Composite Beat the Production Score?

The earlier sections calibrate *where to draw the line* on the existing AI score. This section asks a different question: can the underlying score itself be improved? The production `overall_score` is a weighted average across all 5 rubric dimensions (team, objective, strategy, advantages, feasibility) -- but the earlier correlation study found that **team and objective show no significant relationship** with real Kickstarter outcomes, while feasibility, strategy, and advantages do. Averaging in two noisy dimensions dilutes the signal in the other three.

Three scoring methods were tested head-to-head, all evaluated with the same honest 5-fold cross-validation used throughout this report (same folds, same seed):

- **Method A -- overall_score:** the unmodified production score (baseline).
- **Method B -- simple composite (naive):** a plain, unweighted average of just feasibility + strategy + advantages.
Team/objective were dropped using a correlation study computed on the *full* 141-proposal dataset -- fast to compute, but as shown below, that's a softer form of the same leakage problem this report exists to catch.
- **Method C -- logistic regression:** an L2-regularized model that learns its own weights across all 5 dimensions, fit strictly within each training fold (never seeing its own test fold), producing pooled out-of-fold predicted probabilities.
- **Method B-nested -- the honest version:** the same idea as Method B, but with the "which 3 dimensions?" decision *also* re-made independently within each training fold, exactly like the threshold and the logistic regression weights already were.

Method	ROC-AUC	Avg. Precision	CV Precision	CV Recall	CV F1
A: overall_score (production, 5 dims)	0.7644	0.3611	29.2%	79.2%	0.427
B: simple composite -- naive (dims picked using full dataset)	0.8048	0.4539	35.4%	70.8%	0.4722
C: logistic regression (5 dims, CV-fit)	0.7682	0.3704	32.2%	79.2%	0.4578
B-nested: simple composite -- honest (dims + threshold both fit per-fold)	0.7731	—	32.7%	70.8%	0.4473



The naive number was inflated. Method B looked strong at first (ROC-AUC 0.805) because picking "feasibility, strategy, advantages" using the entire dataset gives the composite a small, undisclosed head start. Once dimension selection is done honestly -- fit per fold, like everything else in this report -- the real number is **ROC-AUC 0.773**: still directionally better than the production score (0.764), but a much smaller edge than first reported.

Is the improvement real, or just luck?

With only 24 successful campaigns to test against, a difference that looks real could still be sampling noise. To check, each pairwise AUC difference was bootstrapped 10,000 times (resample the 141 proposals with replacement, recompute both AUCs, repeat) to get a confidence interval on the gap.

Comparison (AUC difference)	Mean diff	95% CI	P(2nd method better)	Statistically significant?
B (naive) vs. A	+0.0406	[-0.0072, +0.0912]	95.1%	no
B (naive) vs. C	+0.0367	[-0.0074, +0.0816]	95.1%	no
C vs. A	+0.0036	[-0.0190, +0.0288]	60.3%	no

In simple terms: none of the three comparisons reach conventional statistical significance -- every 95% confidence interval includes zero, just barely. Method B does look better than A about 95% of the time across resamples, which is suggestive of a real effect, but it falls just short of the standard bar for "proven." The honest conclusion is **"a promising, directionally consistent improvement that has not yet been shown to be more than chance"** -- not a confirmed result.

Did picking those 3 dimensions "cheat" by looking at all the data?

Separately, the dimension-selection step itself was re-run using *only* each training fold's data, to check whether "feasibility, strategy, advantages" is a stable choice or an artifact of looking at the full dataset at once.

Fold	Top-3 dimensions (by train-fold correlation)	Matches feasibility/strategy/advantages?
Fold 1	advantages, feasibility, strategy	yes
Fold 2	advantages, feasibility, strategy	yes
Fold 3	advantages, feasibility, strategy	yes
Fold 4	feasibility, strategy, team	no
Fold 5	advantages, feasibility, strategy	yes

In simple terms: 4 of 5 folds independently picked the exact same three dimensions using only their own training data -- reassuring stability, not a full-dataset fluke. The one exception (fold 4) swapped "advantages" for "team," showing the choice isn't perfectly rock-solid at the margin either.

Why the learned model didn't win

The logistic regression's own coefficients (fit on the full dataset, shown below for interpretation only -- not used in the evaluation above) explain why: it does lean more on feasibility and strategy, but L2 regularization keeps every dimension's weight in a similar, modest range rather than sharply zeroing out team and objective.

Dimension	Learned weight
Feasibility	+0.1069
Strategy	+0.0797
Team	+0.0402
Advantages	+0.0395
Objective	+0.0317

In simple terms: with only 24 real successes to learn from (about 19 per training fold), there isn't enough data for the model to become *confident* that team and objective are irrelevant, so it hedges and keeps a little weight on everything. The nested version of Method B avoids that hedging problem differently -- not by using outside knowledge for free, but by re-deriving "which dimensions matter" from scratch on each training fold and checking it's stable (Section 5 above: 4/5 folds agree). That's a fair comparison to Method C's per-fold-fit weights, and it still comes out slightly ahead (0.773 vs 0.768) -- a small, honestly-earned edge, worth stating plainly as **a modest, not-yet-statistically-confirmed sign that domain-informed feature selection may generalize better than unconstrained learning in this small-sample regime.**

6. Positive and Negative Outcomes (Updated)

Framed for a research paper, incorporating this calibration study alongside the correlation results from the earlier Kickstarter report.

Positive outcomes

- The AI score has genuine, threshold-independent discriminative power: **ROC-AUC = 0.764** on 141 real campaigns — well above the 0.5 random baseline.
- A properly cross-validated, out-of-sample recalibration lifts recall on real successes from **0% to 58.3%**, and precision from undefined to **26.4%** (a $\sim 1.55\times$ enrichment over the 17.0% base rate) — using held-out test folds only, never in-sample.
- The recalibrated score threshold ($\tau_r \approx 0.55\text{--}0.61$) is stable across all 5 cross-validation folds, suggesting it reflects a real property of the score distribution rather than noise in one particular data split.
- This directly demonstrates that the earlier "0% recall" finding was a **threshold transfer problem**, not evidence the pipeline's scoring is uninformative for this domain — an important, correctable distinction for the paper's narrative.
- Dropping the two dimensions independently shown to lack predictive validity (team, objective) and simply averaging the other three raises ROC-AUC from 0.764 to **0.773**, evaluated fully out-of-sample with dimension selection re-derived per fold (not the inflated 0.805 a naive full-dataset selection would suggest) — a real, if modest, directional edge over both the production score and a learned logistic-regression alternative (Section 5).
- The dimension selection behind that improvement is stable: 4 of 5 independent training folds picked the same three dimensions on their own, without seeing the full dataset.

Negative outcomes / limitations

- Precision remains modest (26.4%) — roughly 3 in 4 campaigns flagged as likely-successful still fail. This is not a strong enough signal to automate a go/no-go decision; it is, at best, a triage aid.
- The gap between in-sample ($F1=0.478$) and held-out ($F1=0.364$) performance is a clean, quotable illustration of overfitting risk — and a reason to be skeptical of any threshold reported without a train/test split, including the biotech dataset's own OR-logic threshold, which was calibrated in-sample on all 12 proposals with no held-out check.
- Fold-level F1 is noisy (0.0–0.47) because each held-out fold contains only ~ 5 real successes — the aggregated result is more trustworthy than any individual fold, but the overall sample is still small for the positive class, and confidence intervals on precision/recall would be wide.
- This calibration is specific to Kickstarter's score distribution; it should not be assumed to transfer to a third domain any more than the original biotech threshold transferred to Kickstarter.
- The composite-score improvement (Section 5) does **not** reach statistical significance: a 10,000-sample bootstrap on the ROC-AUC gap gives a 95% CI of $[-0.0072, +0.0912]$, which includes zero. It should be reported as a promising, directionally consistent signal ($\sim 95\%$ bootstrap probability of being real) — not a confirmed improvement — until tested on a larger sample.

7. Interesting Findings Worth Highlighting in a Paper

- **Domain-shift threshold failure, precisely diagnosed and fixed:** a threshold transferred unmodified across domains produced 0% recall; recalibrating the same decision rule (same OR-logic shape, new cutoffs) on the target domain, validated out-of-sample, restored 58.3% recall — a clean before/after story about covariate shift in LLM-based scoring rubrics.
- **Threshold-free vs. threshold-based evidence tell a consistent story:** ROC-AUC (0.764, threshold-independent) and the earlier Spearman correlation (0.366, rank-based) both independently support that the score carries real signal — while the naive classification accuracy (83%, driven by base rate) and the original 0% recall both independently misled in the opposite direction. Worth discussing as a methods lesson: which metrics are trustworthy under class imbalance and domain shift, and which are not.
- **An honest overfitting demonstration, in-repo:** the in-sample calibration ($F1=0.478$) vs. held-out cross-validation ($F1=0.364$) gap is a small, self-contained illustration of exactly the failure mode the paper should be careful to avoid elsewhere — including, arguably, in its own existing 12-proposal biotech threshold calibration, which was never validated out-of-sample.
- **Small positive-class sizes make single train/test splits unreliable:** with only 24 successes in 141 proposals, stratified k-fold cross-validation was necessary to get a stable estimate at all — a methodological point that generalizes to any future outcome-based validation this pipeline undergoes on similarly rare-success datasets.
- **A quiet leak, caught by nesting the whole pipeline:** a "zero-fitting" composite score that simply dropped two known-irrelevant dimensions still turned out to be quietly leaking information -- its feature-selection step used the full dataset's correlations, inflating ROC-AUC from an honest 0.773 to an overstated 0.805. Re-deriving the dimension choice independently within each fold (4/5 folds agreed) recovered a defensible, smaller number. A concrete illustration that leakage doesn't require model fitting to sneak in -- even a one-time, seemingly "free" data-driven design choice can leak if it touches the evaluation set.
- **A modest, not-yet-significant improvement, reported honestly:** the fully-nested composite score beats the production score on point estimate (ROC-AUC 0.773 vs 0.764) with a ~95% bootstrap probability of being a real effect, but the 95% confidence interval on the difference still includes zero at this sample size. Presenting a result at exactly this level of confidence, rather than rounding up to "proven," is itself a useful methodological example for the paper's broader trustworthiness argument.

Glossary: Train/Test and Related Terms (Plain-Language)

Train set / test set

The train set is the data used to pick the threshold. The test set is different data, held back, used only to check how well that threshold actually performs. Using the same data for both makes results look better than they really are.

K-fold cross-validation

Splits the data into k equal groups (folds). Repeats the train/test process k times, each time using a different fold as the test set and the rest as training. Every data point gets tested exactly once, using a threshold it never influenced.

Stratified

When splitting into folds, we made sure each fold has roughly the same proportion of successes as the full dataset (~17%), so no fold ends up with zero or almost-zero successes to test against.

Overfitting

When a threshold (or model) is tuned so precisely to one specific dataset that it stops generalizing to new data. The in-sample vs. held-out gap in this report (F1 0.478 vs 0.364) is a direct, measured example of it.

ROC-AUC

The probability that a randomly chosen success scores higher than a randomly chosen failure. 0.5 = no better than random guessing; 1.0 = perfect separation. Does not depend on any specific threshold.

Precision vs. recall

Precision: of the campaigns flagged as "likely to succeed," what fraction actually did? Recall: of the campaigns that actually succeeded, what fraction did we flag? There's usually a trade-off between the two.

F1 score

A single number combining precision and recall (their harmonic mean). Used instead of accuracy here because accuracy is misleading when one outcome (failure) is much more common than the other (success).

Base rate

How often the positive outcome happens by default — here, 17.0% of Kickstarter campaigns succeed. Any useful signal has to beat this baseline, not just look impressive in isolation.

Source data: `results/KickstarterCalibrated/evaluation/calibration_results.json` (generated by `calibrate_kickstarter_threshold.py`). Report generated by `generate_kickstarter_calibration_report.py`. Random seed 42, 5-fold stratified cross-validation.